

EduGenie: Google Gemini Powered Learning Assistant

7. Project Documentation

This document organizes the EduGenie project's documented technical and user-facing information into a formal project documentation record suitable for the repository's Project Documentation section.

7.1 Project Overview

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning using generative AI.

It supports Q&A, simplified explanations, quiz generation, personalized recommendations, and summarization.

7.2 Technology Stack

Backend: FastAPI.

Frontend: HTML and CSS.

AI: Gemini 1.5 Pro via API and LaMini-Flan-T5-783M locally.

Server: Uvicorn.

Templating: Jinja2.

Runtime prerequisite: Python 3.10+.

7.3 Repository Documentation

main.py

explanation_module.py

qna.py

quiz_module.py

summary_module.py

learning_path.py

templates/index.html

static/style.css

requirements.txt

7.4 API Documentation

/qa — question answering.

/explain — concept explanation.

/quiz — quiz generation.

/summarize — summarization.

/learn/recommendations — learning recommendations.

7.5 User Workflow

The user chooses a task from the dropdown.

The user enters educational text or a question.

The user submits the form.

The frontend sends a POST request to FastAPI.

The backend invokes the corresponding module.

The result is shown below the input.

7.6 Deployment Documentation

The documented local command is `uvicorn main:app --reload`.

The documented local address is `http://127.0.0.1:8000`.

7.7 Documented Outcomes

The project supports dynamic and interactive learning.

The modular architecture allows future upgrades.

The source identifies future possibilities including voice interaction, multilingual support, mobile applications, progress dashboards, gamification, adaptive learning, group study, teacher/parent dashboards, LMS integration, image/PDF input, real-time synchronization, and smart notifications.

7.8 Project Metadata

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Conclusion

The documentation consolidates the project's purpose, architecture, modules, API endpoints, frontend flow, deployment method, and documented future scope into a single reference for the repository's Project Documentation stage.

Source basis: EduGenie project document supplied with this request and the repository structure shown in the supplied reference image. Where the source did not specify an implementation detail, this documentation preserves the documented scope rather than inventing project facts.